

Q7-18

编题34 20230804

思路: (6 km)

P3 409 空气物性

可得 (比值)

 U_∞

$$p_\infty = 47.217 \text{ kPa}$$

$$T_\infty = 249.250 \text{ K}$$

$$\rho_\infty = 0.659 \text{ kg/m}^3$$

$$U_\infty = 2.4167 \times 10^{-5} \text{ m/s}$$

$$M_1 = 2.0$$

(E) P-M 波.

$$\therefore U(M_2) - U(M_1) = 5$$

$$U(M_2) - U(2.0) = 10$$

$$\therefore U(M_2) = 26.5 + 10 = 36.5$$

反查
P3 424

$$\left(M_2 = 2.390 \right) \quad \frac{p_2}{p_1} = \left(\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right)^{\frac{\gamma}{\gamma-1}} = 0.5436, \therefore p_{\text{upper}} = 25.667 \text{ kPa}$$

(F)

斜激波

$$\left(\text{PM} \right) \quad \frac{T_2}{T_1} = \frac{T_2}{T_0} \cdot \frac{T_0}{T_\infty} = (0.46676) \cdot \left(\frac{1}{0.5556} \right) U_2 = M_2 a_2 = M_2 \sqrt{\gamma R T_2} = 694.048 \text{ m/s}$$

$$\therefore (8 - \beta - Ma) \text{ 查表得 } \beta = 39.5^\circ, \therefore T_2 = (0.8602)(T_\infty) = 209.821 \text{ K}$$

$$\therefore M_{1n} = M_1 \sin \beta = 1.272, \text{ 查表 } M_{2n} = 0.80165, M_2 = \frac{M_{2n}}{\sin(\beta - \delta)}$$

$$\text{代入 } \frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_{1n}^2 - 1) = 1.721 \quad \approx 1.628$$

$$\therefore p_2 = 81.260 \text{ kPa}, \quad p_{\text{lower}} = p_2 - 81.260 \text{ kPa} \quad U_3 = M_3 a_3$$

$$\therefore U_\infty = M_1 a_\infty, \quad a_\infty = \sqrt{\gamma R T_\infty} = \sqrt{(1.4)(287)(249.25)} \approx 316.463 \text{ m/s}$$

$$\therefore U_\infty = 632.925 \text{ m/s}$$

$$\frac{T_3}{T_\infty} = 1.4088, \quad T_3 = 351.143$$

(2)

$$C_L = \frac{F_L}{\frac{1}{2} \rho_\infty U_\infty^2 c}$$

$$F_L = \Delta p (c \cdot 1)$$

$$\Delta p = (p_{\text{upper}} - p_{\text{lower}})$$

$$= 55.593 \text{ kPa}$$

$$= 4.2117 \times 10^{-1}$$

$$\approx 0.421$$

$$C_D = \frac{F_D}{\frac{1}{2} \rho_\infty U_\infty^2 c}$$

$$F_D = F_L \tan \alpha$$

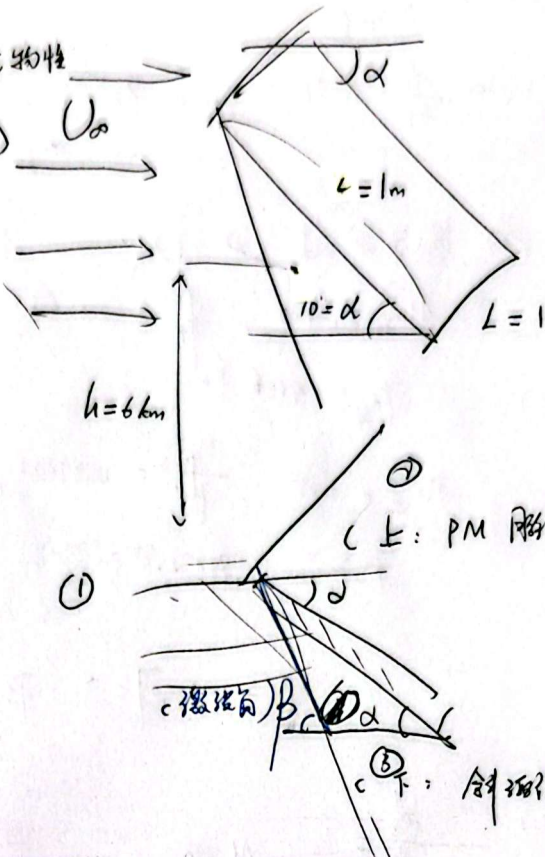
$$\therefore C_D = C_L \tan \alpha$$

$$\approx 0.0743$$

$$(3) \quad \dot{I} = c(U_3 - U_2)$$

$$= 1(611.517 - 694.048)$$

$$= -82.531 \text{ m/s}$$



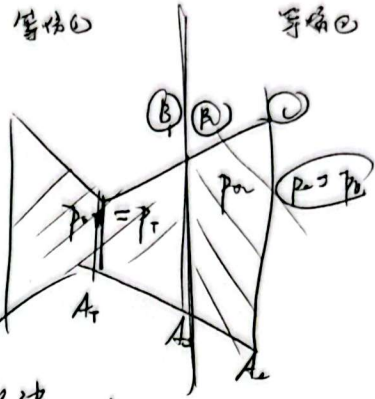
Q7.20 $\frac{A_c}{A^*} = 11.91$, $p_{01} = 7 \times 10^5 \text{ Pa}$, $T_{01} = 500 \text{ K}$, $p_b = 2.2 \times 10^5$

Step ① 先假设激波存在
(1) $(p_{020} < p_b < p_{021})$

$\text{Ma} > 1$
 $\text{Ma} = 4.0$ $\frac{p_{021}}{p_{01}} = 0.00658$, $p_{01} = p_{02}$ $\therefore p_{021} = 4666 \text{ Pa}$

假设
在管嘴出口

$\text{Ma} < 1$
 $\text{Ma} = 0.205$ $\frac{p_{021}}{p_{01}} = 0.99825$, $p_{01} = p_{02}$ $\therefore p_{021} = 6.998 \times 10^5 \text{ Pa}$



(2) $q(\text{Ma})$

(2) $\frac{1}{\text{Ma}} \left[\frac{2 + (\gamma - 1)\text{Ma}^2}{\gamma + 1} \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} = \frac{A_0}{A^*} \frac{p_0}{p_{01}} \sqrt{\frac{T_{01}}{T_0}}$ $\therefore p_0 = p_b$ 出口(亚音速)

$\text{Ma} = 0.15016$, $\text{Ma} \rightarrow \text{Ma}$ 等值, $\frac{A}{A^*} = \frac{A_c}{A_b} = 3.9103$

$\text{Ma} \rightarrow \text{Ma}$ 等值

$\text{Ma} \rightarrow \text{Ma}$ 等值: $\therefore \text{Ma} = 0.15016$

方法① $\frac{p_{02}}{p_{01}} = \left(\frac{p_{02}}{p_b} \right) \left(\frac{p_b}{p_{01}} \right)$

$= \left(\frac{1}{0.98244} \right) \left(\frac{2.2 \times 10^5}{7 \times 10^5} \right) = 0.3280$

$\frac{p_{02}}{p_b} = \left(1 - \frac{\gamma - 1}{2} \text{Ma}^2 \right)^{\frac{\gamma}{\gamma - 1}}$
 $= 1.0159$

$\therefore p_{02} = 0.3280 p_{01} = 2.296 \times 10^5 \text{ Pa}$

$\frac{A_3}{A^*} = 0.32834$ #

$M_1 = 2.0$
 $M_2 = 0.47519$

$M_{asf} = M_1 = 2.0$ #

(3) $\text{Ma}_0 = 0.15016$ #

$\frac{T_{02}}{T_c} = \frac{T_{01}}{T_c} = 1 + \left(\frac{\gamma - 1}{2} \right) \text{Ma}^2$

$\frac{T_{01}}{T_c} = \frac{500}{T_c} = 1.0044$ $\therefore T_c = 497.76 \text{ K}$ #

Q8.4 (已知) Q, d, τ_w, ρ, μ 求 μ

$\therefore \textcircled{1} v = \frac{\mu}{\rho}$ 題: 粘性係數 (μ)

$\therefore \tau = \mu \left| \frac{du}{dr} \right|$

$u(r) = u_{max} \left[1 - \left(\frac{r}{R} \right)^2 \right], \therefore \frac{du}{dr} = -\frac{2u_{max} \cdot r}{R^2}, \text{ 在 } r=R \text{ 處 } \frac{du}{dr} = -\frac{2u_{max}}{R}$

$\therefore \tau_w = \mu \left(\frac{2u_{max}}{R} \right) = \mu \left(\frac{2 \cdot \frac{Q}{A}}{R} \right) = \frac{8\mu Q}{d^3}$

$\therefore \mu = \frac{\tau_w d^3}{8Q}, \quad v = \frac{Q}{A}$
 $= \frac{(48)(0.075)^3}{8 \left(\frac{0.007}{\pi \left(\frac{0.025}{2} \right)^2} \right)}$

$\approx 0.284 \text{ Pa}\cdot\text{s}$

$\textcircled{2} R_e = \frac{\rho v d}{\mu} = 335 \text{ (非層流)}$

Q8.5

$d = 1.5 \text{ mm}, \quad m = 13.7 \text{ mg}$

$\rho_{oil} = 0.85 \rho_w, \quad \mu = ?$

$\therefore F_d (\text{Stokes}) = 3\pi\mu d v$

$mg = F_b + F_d \text{ (平衡)}$

$mg = \rho_{oil} g \left(\frac{4}{3}\pi R^3 \right) + F_d$

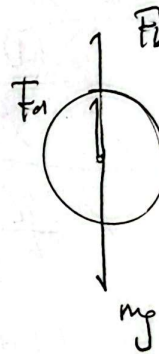
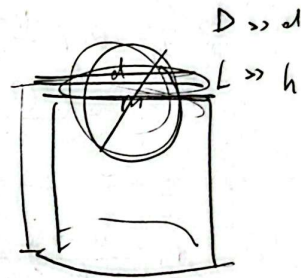
$\therefore F_d = 1.178 \times 10^{-4} \text{ N}$

$\mu = \frac{F_d}{3\pi d v}, \quad v = \frac{h}{t}$

$= \frac{1.178 \times 10^{-4}}{3(\pi)(1.5 \times 10^{-3}) \left(\frac{0.007}{5} \right)}$

$\approx 0.933 \text{ Pa}\cdot\text{s}$

~~Stokes = h~~
(勾選)



0.6 $\frac{dp}{dx} = 0$ (全流场为常数)

$g_x = 0$ 对 $u(y)$ - τ_{xy} $(h+h_2)$

$\tau_1 = \mu_1 \left(\frac{u_0 - 0}{h_1} \right)$ $\because \mu \left(\frac{du}{dy} \right)$
 $\tau_2 = \mu_2 \left(\frac{U - u_0}{h_2} \right)$

$\tau_1 = \tau_2$

$\mu_1 \left(\frac{u_0 - 0}{h_1} \right) = \mu_2 \left(\frac{U - u_0}{h_2} \right)$

$u_0 \left(\frac{\mu_1}{h_1} + \frac{\mu_2}{h_2} \right) = \frac{\mu_2 U}{h_2}$

$u_0 = \frac{\mu_2 h_1 U}{\mu_1 h_2 + \mu_2 h_1}$

u_0 - 为解界面速度

$u(y) = \frac{u_0}{h_1} y + u_0 + \frac{U - u_0}{h_2} (y - h_1)$
 $\tau = \frac{\mu_1 \mu_2 U}{\mu_1 h_2 + \mu_2 h_1}$

$\frac{du}{dx} = 0 \Rightarrow \frac{du}{dy} = 0$
 $U = \text{const.}$

